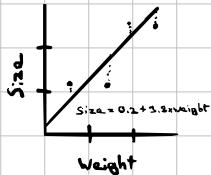


Regularization

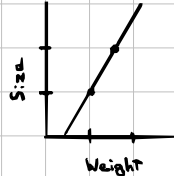
Ridge regression (L2 regularization)

- Introduce bias in training for less variance in testing.
- Regularization = Desensitization

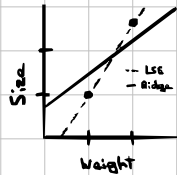
Linear regression



With a lot of data
min of LSE is good
for predictions



With two data points
min of sum of LSE
is 0, but clear overfit



With Ridge regression
minimize:
 $\sum \text{squared residuals}$
 $+ \underbrace{\lambda \times \text{slope}^2}_{\text{penalty}}$



Increasing λ to desensitize
size to weight.

A: $\lambda = 1$, B: $\lambda = 10$, C: $\lambda = 10000$

Complex models

We can use Ridge regression on multiple variables, say:

$$\text{size} = y\text{-intercept} + a \times \text{Weight} + b \times \text{Diet} + c \times \text{Age}$$

Then minimize: $\sum (y - \hat{y}_i)^2 + \lambda (a^2 + b^2 + c^2)$

In general, all variables apart from y-intercept: $\min_{\beta} \sum (y - \hat{y})^2 + \lambda \sum_j \beta_j^2$

Lasso regression (L1 regularization)

- Lasso same idea as Ridge with goal: $\min_{\beta} \sum (y - \hat{y})^2 + \lambda \sum_j |\beta_j|$

Difference: L2 vs. L1 regularization

L2 will penalize small coefficients a little:

$$(\text{big}) \beta_1 = 0.8 \rightarrow \beta_1^2 = 0.64 \quad (-0.36)$$

$$(\text{small}) \beta_2 = 0.1 \rightarrow \beta_2^2 = 0.01 \quad (-0.09)$$

Ridge will prefer to penalize big (most explanatory) coefficient

L1 can eliminate useless variable:

$$\beta_1 = 0.8 \rightarrow |\beta_1| = 0.8$$

$$\beta_2 = 0.1 \rightarrow |\beta_2| = 0.1$$

So Lasso could eliminate a variable, $\beta_j = 0$.